

Equation from Intercepts

Practice Sheet 1 - GCSE Maths (Higher)

Name: _____

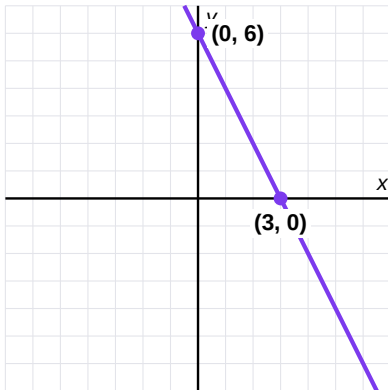
Date: _____

Each line crosses the axes at the two points shown. Write its equation in the form $y = mx + c$.
Remember: c is where it crosses the y -axis, and $m = \text{change in } y / \text{change in } x$.

Score: _____ / 24

Section A: Whole-number gradients (2 marks each)

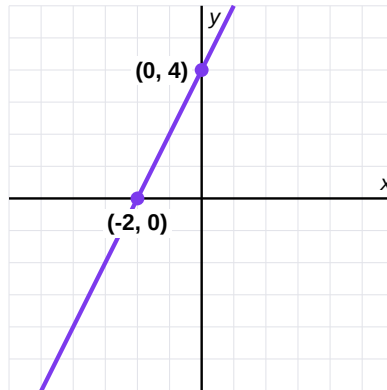
A1 Crosses at (0, 6) and (3, 0)



Working:

$y =$ _____

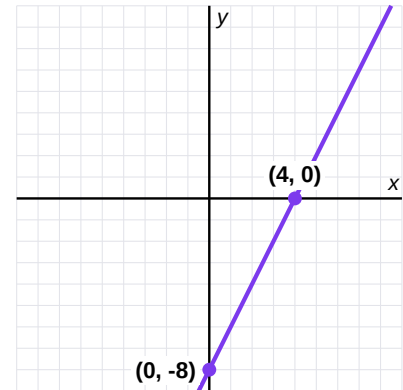
A2 Crosses at (0, 4) and (-2, 0)



Working:

$y =$ _____

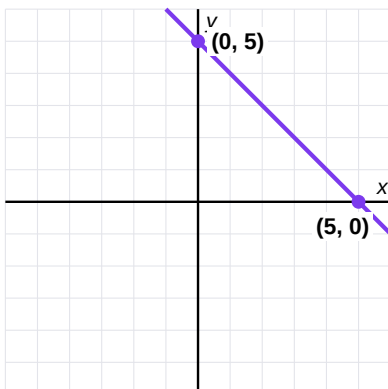
A3 Crosses at (0, -8) and (4, 0)



Working:

$y =$ _____

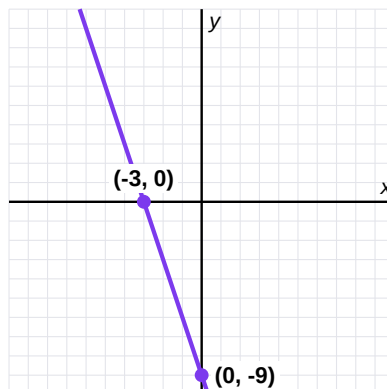
A4 Crosses at (0, 5) and (5, 0)



Working:

$y =$ _____

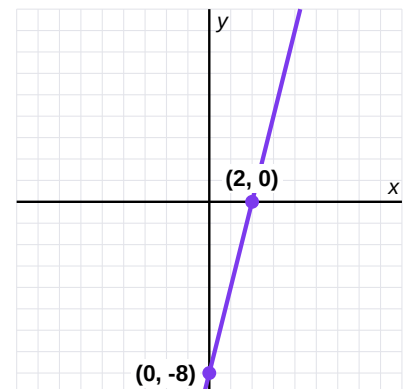
A5 Crosses at (0, -9) and (-3, 0)



Working:

$y =$ _____

A6 Crosses at (0, -8) and (2, 0)



Working:

$y =$ _____

Equation from Intercepts

Practice Sheet 1 - GCSE Maths (Higher)

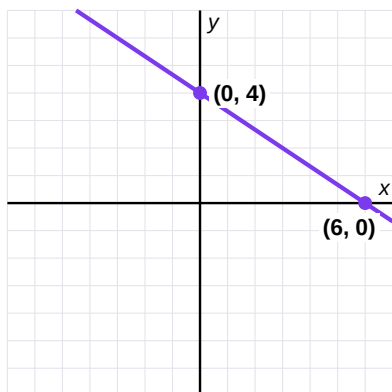
Name: _____

Date: _____

Leave fractions as fractions (not decimals). Q5 and Q6 want the form $ax + by = c$, where a , b , c are integers.

Section B: Fraction gradients - Higher (2 marks each)

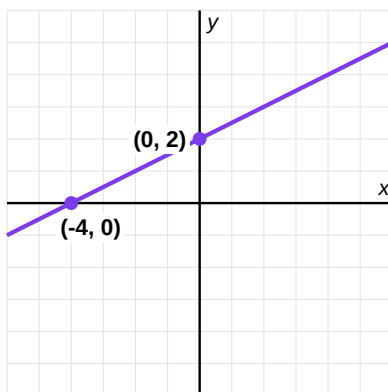
B1 Crosses at $(0, 4)$ and $(6, 0)$



Working:

$y =$ _____

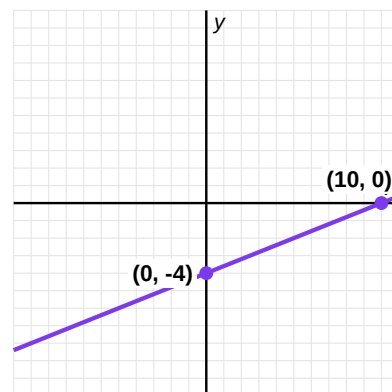
B2 Crosses at $(0, 2)$ and $(-4, 0)$



Working:

$y =$ _____

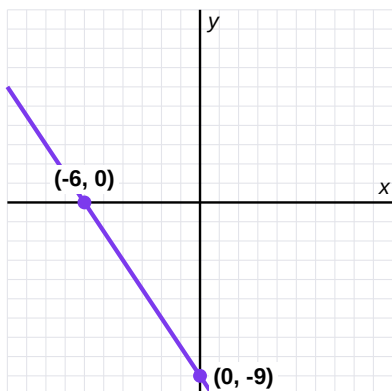
B3 Crosses at $(0, -4)$ and $(10, 0)$



Working:

$y =$ _____

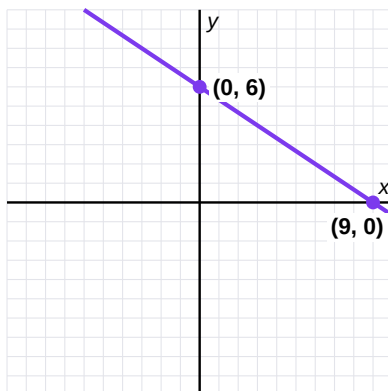
B4 Crosses at $(0, -9)$ and $(-6, 0)$



Working:

$y =$ _____

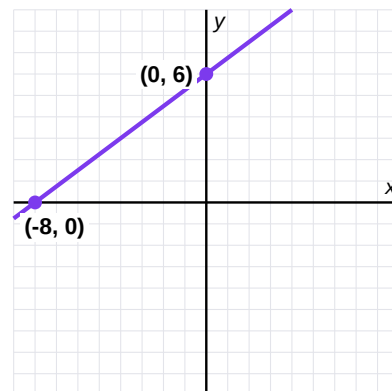
B5 Crosses at $(0, 6)$ and $(9, 0)$



Working:

$ax + by = c:$ _____

B6 Crosses at $(0, 6)$ and $(-8, 0)$



Working:

$ax + by = c:$ _____